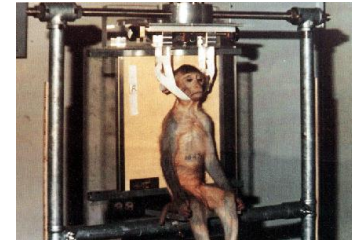


Assessing and Managing Risk in Science & Engineering

- 1st dimension: objective risk v. subjective uncertainty
- 2nd dimension: quantitative vs. qualitative descriptions
- quantitative-objective: EV; quantitative-subjective EUT
- qualitative
 - quantification attempts: objective – intervals; subjective - PIR
 - worst case
 - robustness
 - symmetry
 - precaution
- Simple safety principles: inherent safety, safety factors, and multiple barriers.
- risk exposure – distributional issues

1. Risk in Science & Engineering

- Science and technology has impacts on the world
 - often has offered improvements, but arguably it sometimes also has made things worse
- for all, including scientist s& engineers: future outcomes uncertain -> RISK



- Scientists and engineers often need to *identify* and *assess* risks
- Information from science and engineering is often used to *manage* risks: to *eliminate* or *reduce them*.
- Scientific results and new technology sometimes *create* new risks

Now I've used the word "risk" as something that can be created but is undesirable – and which needs to be assessed and managed

1. Def: Risk =_{df} *an unwanted event which may or may not occur.*

e.g. "Damage to lungs due to inhalation of smoke is a big risk in fires"

Now I've used the word "risk" as something that can be created but is undesirable – and which needs to be assessed and managed

1. Def: Risk =_{df} *an unwanted event which may or may not occur.*

2. Def: Risk =_{df} *the cause of an unwanted event which may or may not occur.*

e.g. "Smoking is the most important health-risk in the developed countries"

Now I've used the word "risk" as something that can be created but is undesirable – and which needs to be assessed and managed

1. Def: Risk =_{df} an *unwanted* event which may or may not occur

2. Def: Risk =_{df} the *cause* of an unwanted event which may or may not occur

Primary prevention:

eliminating the *cause* of an unwanted event.

E.g. preventing fire hazards by using only inflammable materials

Secondary prevention:

providing means of mitigating the unwanted consequences of such a 'risk' cause.

E.g. mitigating fire hazards by installing fire extinguishers.

Now I've used the word "risk" as something that can be created but is undesirable – and which needs to be assessed and managed

1. Def: Risk =_{df} *an unwanted event which may or may not occur*

2. Def: Risk =_{df} *the cause of an event which may or may not occur*

Problems with Secondary prevention: Boeing 737

Boeing's engineers developed a control system called MCAS - Manoeuvring Characteristics Augmentation System - to mitigate the risk of stalling from pilot error.

As we now know: this can introduce additional risks of its own!



Now I've used the word "risk" as something that can be created but is undesirable – and which needs to be assessed and managed

1. Def: Risk =_{df} *an unwanted event which may or may not occur*

2. Def: Risk =_{df} *the cause of an unwanted event which may or may not occur*

another implication of difference:

* unwanted events that are always bad (nobody wants lung damage)

* causes with bad effects might have other **good** effects (e.g. enjoy smoking cigarettes)



when banks offer investment opportunities, they describe them in terms of their risk

That would not make any sense if risk was only a bad instead, 'risk' here describes the uncertainty which of either positive or negative outcomes might occur

3. Def: Risk =_{df} the spread of possible (good & bad) outcomes that might occur.



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That would not make any sense if risk was only a bad instead, 'risk' here describes the uncertainty which of either positive or negative outcomes might occur

3. Def: Risk =_{df} the spread of possible (good & bad) outcomes that might occur.



Qualitative notions of risk.
Neither severity nor
probability is quantified.

when banks offer investment opportunities, they describe them in terms of their risk

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Quantitative Notions of Risk

4. Def: Risk =_{df} the *probability* of an unwanted event which may or may not occur.

e.g. "Smokers risk dying of lung cancer with probability 0.014, non-smokers with 0.001"

Quantitative Notions of Risk

4. Def: Risk =_{df} the *probability* of an unwanted event which may or may not occur.

e.g. "Smokers risk dying of lung cancer with probability 0.014, non-smokers with 0.001"

5. Def: Risk =_{df} the *severity* of an unwanted event which may or may not occur.

e.g. "Crash victims not seat-belted on average stay 4.3 days longer in hospital than those with seatbelts fastened"

Quantitative Notions of Risk

4. Def: Risk =_{df} the *probability* of an event which may or may not occur

e.g. "Smokers risk dying of lung cancer with 0.001"

5. Def: Risk =_{df} the *severity* of an event which may or may not occur

e.g. "Crash victims not seat-belted on average die in hospital faster than those with seatbelts fastened"

Complication – objective or subjective?

Probabilities

objective frequencies – how many of all smokers died of lung cancer?

subjective estimate – how much would you pay for an insurance cover of \$100k against lung cancer, if you are a smoker? \$20 if 2%, \$10k if 1%...

Quantitative Notions of Risk

4. Def: Risk =_{df} the *probability* of an event which may or may not occur

e.g. "Smokers risk dying of lung cancer with 0.001"

5. Def: Risk =_{df} the *severity* of an event which may or may not occur

e.g. "Crash victims not seat-belted on average stay 4.3 days longer in hospital than those with seatbelts fastened"

Complication – objective or subjective?

Severities

objective values like money, days in hospitals, number of people killed,...

subjective evaluations – how much is \$20 worth to you if you are starving?

Quantitative Notions - Knowledge requirements

when describing risk quantitatively, we need to know quite a lot:

- what possible outcomes an action might have
- with what probability each outcome might occur

we can distinguish situations where we do not have this knowledge from those where we do....

Decision making under certainty	Action has only one outcome
Decision making under risk	Action has more than one possible outcomes ; results of each outcome is known; probability of each outcome is known .
Decision making under ignorance	Action has more than one possible outcomes; each outcome is known; probabilities of some outcomes are not known .
Decision making under deep uncertainty	Action has more than one possible outcomes; not all of these outcomes are known , nor are their probabilities.

“there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns – the ones we don't know we don't know.”

Donald Rumsfeld (*1932)



Decision making
under **ignorance**

Action has more than one possible outcome; each outcome is known; **probabilities** of some outcomes are **not known**.

Decision making
under **deep
uncertainty**

Action has more than one possible outcomes; **not all of these outcomes are known**, nor are their probabilities.

Different approaches in different disciplines

Nuclear engineering: predominantly “risk-based approach” as the use of probabilities to characterize hazards and prioritize their abatement.

Software engineering: their risks:

- errors due to unusual or unforeseen inputs
- intrusion by hackers
- operator mistakes caused by confusing human-machine interfaces

However, hardly ever quantified. Instead: scenarios with undesired event or event chain rather than a probability

Civil engineering, their risks:

- structural failures, such as a collapsing house, bridge, or dam
- their probability is very low – and it is generally agreed that such events should be avoided at almost any price
- therefore: don't calculate numerical probabilities of such failures
- Instead, use rules of thumb, including numerical **safety factors**, to ensure avoidance of such events

In **climate science:** multiple model set

- first quantitative: calculate mean temperature increase from different models
- second non-quantified: determine spread of projections, from different models
- newest proposals: focus on individual scenarios instead

SUMMARY – different dimensions of risk

- causes vs. effects
- qualitative vs. quantitative
- objective vs. subjective
- qualitative differences about what is unknown

2. Expected Value Models of Risk

The expected value of a discrete random variable is the probability-weighted average of all its possible values.

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Random variable – variable whose possible values are numerical outcomes of a random phenomenon

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Example:

Lottery ticket A yields \$100 with probability 0.4, \$500 with probability 0.2 and nothing otherwise

Lottery ticket B yields \$75 with probability 0.3, \$750 with probability 0.1, and \$2 with probability 0.6

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Lottery ticket A yields \$100 with probability 0.4, \$500 with probability 0.2 and nothing otherwise

$$EV(A) = \$100 \cdot 0.4 + \$500 \cdot 0.2 = \textbf{\$140}$$

Lottery ticket B yields \$75 with probability 0.3, \$750 with probability 0.1, and \$2 with probability 0.6

$$EV(B) = \$75 \cdot 0.3 + \$750 \cdot 0.1 + \$2 \cdot 0.6 = \textbf{\$98.7}$$

Risk not just been measured by but identified with EV model.

5. risk = the *statistical expected value* of an uncertain event.

If the numerical measure of the outcome space is a good, then risk decreases with increasing EV:

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Risk not just been measured by but identified with EV model.

5. risk = the *statistical expected value* of an uncertain event.

If the numerical measure of the outcome space is a bad, then risk increases with increasing EV:

Technology A: 200 deaths with probability 0.01, 40 deaths with probability 0.3, no deaths otherwise

$$EV(A) = 200 \cdot 0.01 + 40 \cdot 0.3 = 14$$

Technology B: 150 deaths with probability 0.2, 20 deaths with probability 0.4, 10 deaths with probability 0.4

$$EV(B) = 150 \cdot 0.2 + 20 \cdot 0.4 = 38$$

Arguments for Using EV for Risk Modelling

Allows economic optimization

- if outcomes homogeneous, use their quantities
- if outcomes heterogeneous, find a common currency (e.g. money)
- calculate EV for each choice alternative
- choose alternative with highest monetary EV / lowest 'bad' EV
- determine Risks and benefits are assigned monetary values.

Advantages:

Takes both the good and the bad into considerations and attempts to form a rational ground for a trade-off decision. I.e. answer to question: "is the additional gain in safety worth the effort?"

Contrast: safety factors

no way to translate the difference between using the safety factor 2.0 and the safety factor 3.0 in the design of a bridge into a quantifiable effect on safety. Thus impossible to calculate the marginal cost of risk reduction.

Arguments for Using EV for Risk Modelling

Allows separating risk assessment – risk management

Risk assessment



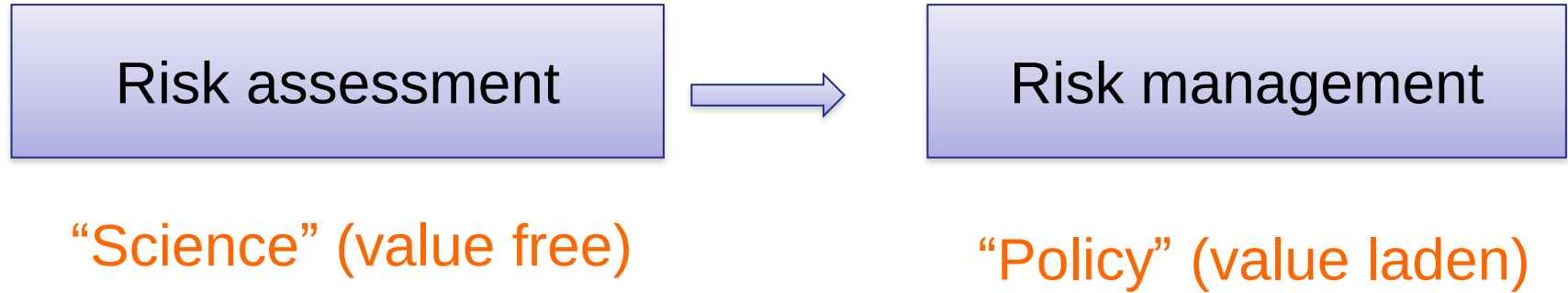
Risk management

Collecting and assessing the relevant information and on this base characterizing the nature and magnitude of the risk.

Combining risk assessment with economical, technological, political and social information. Based on this makes decision on what measures – if any – should be taken to reduce the risk.

Arguments for Using EV for Risk Modelling

Allows separating risk assessment – risk management



Arguments for Using EV for Risk Modelling

Allows separating risk assessment – risk management

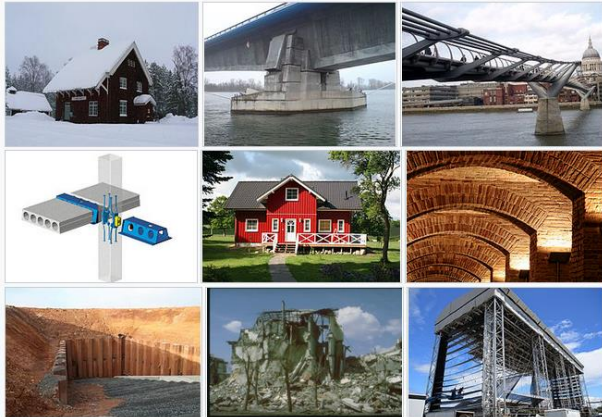
Risk assessment



Risk management

“Science” (value free)

“Policy” (value laden)



Contrast with **Eurocodes**, one of its purposes:

- requirements for mechanical strength and stability and safety in case of fire
- risks assessed by experts
- experts decided what technical specifications needed to avoid such risks (risk management!)
- little non-expert, public involvement in management

Problems of Defining Risk as EV

Technology C: 110 deaths probability 0.5, 90 deaths with probability 0.5

Technology D: 150 deaths with probability 0.4, 100 deaths with probability 0.2, 60 deaths with probability 0.3, 20 deaths with probability 0.1.

Problems of Defining Risk as EV

Technology C: 110 deaths probability 0.5, 90 deaths with probability 0.5

$$EV(C) = 100$$

Technology D: 150 deaths with probability 0.4, 100 deaths with probability 0.2, 60 deaths with probability 0.3, 20 deaths with probability 0.1.

$$EV(D) = 100$$

Problems of Defining Risk as EV

Technology C: 110 deaths probability 0.5, 90 deaths with probability 0.5

$$EV(C) = 100 \qquad \sigma_C = 10$$

Technology D: 150 deaths with probability 0.4, 100 deaths with probability 0.2, 60 deaths with probability 0.3, 20 deaths with probability 0.1.

$$EV(D) = 100 \qquad \sigma_D \approx 46$$

C and D intuitively have different risks, but they have the same EV.
Additional measures needed – like variance here. Don't define risk as EV!

Problems of Defining Risk as EV

Technology C: 110 deaths probability 0.5, 90 deaths with probability 0.5

$$EV(C) = 100 \qquad \sigma_C = 10$$

Technology D: 150 deaths with probability 0.4, 100 deaths with probability 0.2, 60 deaths with probability 0.3, 20 deaths with probability 0.1.

$$EV(D) = 100 \qquad \sigma_D \approx 46$$

Which of these technologies is riskier? Which one should one choose?

-> Depends on one's risk preferences!

Risk Preferences

Risk-loving: putting more weight on potentials gains than potential losses

Risk-averse: putting more weight on potential losses than potential gain

Risk Preferences

Risk-loving: putting more weight on potentials gains than potential losses

Risk-averse: putting more weight on potential losses than potential gains

Most people are risk-averse – "St. Petersburg Paradox", Daniel Bernoulli 1738.

Risk Preferences

Risk-loving: putting more weight on potential gains than potential losses

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Most people are risk-averse – "St. Petersburg Paradox", Daniel Bernoulli 1738.

game offered to you:

- a fair coin is tossed at each stage
- initial stake 2 dollars
- stake doubled every time heads
- game ends as soon as tails appears, player wins latest stakes

Thus the player wins 2 dollars if tails appears on the first toss, 4 on the second, 8 on the third and so on...

how much would you be willing to pay for such a game?

pause video and answer for your self!

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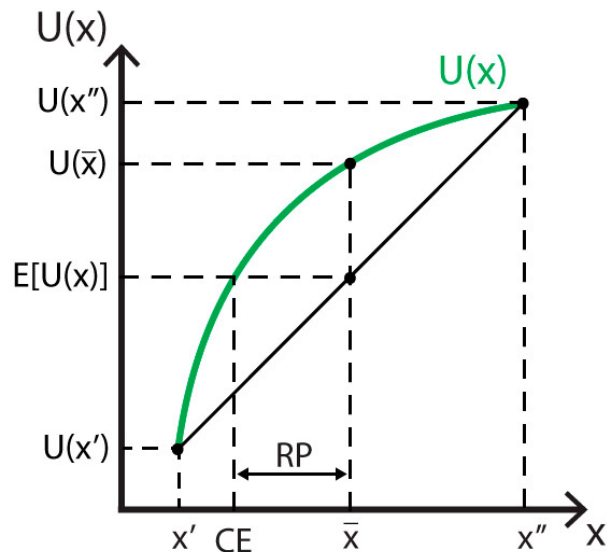
pause video and answer for your self!

EV calculation:

$$\begin{aligned} E &= \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \frac{1}{16} \cdot 16 + \dots \\ &= 1 + 1 + 1 + 1 + \dots \\ &= +\infty. \end{aligned}$$

Bernoulli: that is absurd – nobody would pay all their fortune to play this game. Are they irrational? No – they just value large sums less than small sums

Risk Preferences



Risk averse individual

$$E[U(x)] < U(\bar{x})$$

$$CE < \bar{x}$$

$$0 < A$$

In the St. Petersburg game, individuals value money sums as less the bigger they are. Thus the prospect of winning \$1 Mio with probability 0.01 is much less than having \$10k for certain.

This is formalized by introducing the concept of **utility**. A risk averse individual does not value every dollar the same. Rather, the 1st dollar has a higher utility to her than the 2nd dollar, and the 2nd a higher than the third. This results in the marginal diminishing utility curve.

For evaluation of uncertain outcomes, this is hugely important: the expected **utilities** of the uncertain outcomes $E(U(x))$ is **less** than the utility of the expected value $U(\bar{x})$.

Such a player will value a gamble as less than the gambles expected value (as we saw in St. Petersburg game)

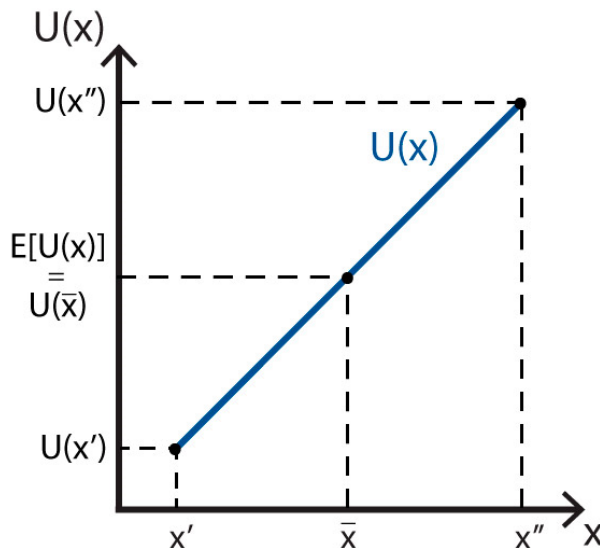
x)

x

Risk Preferences

But this is a **subjective** parameter – some people are risk averse, others are not.

Those evaluating in accord with EV are called risk neutral

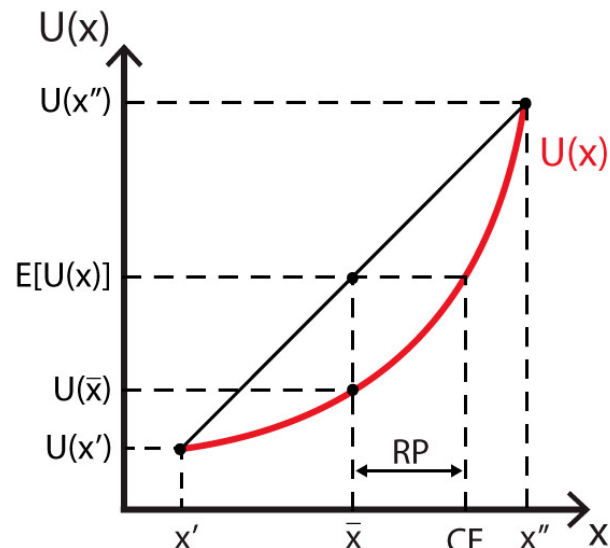


Risk neutral individual

$$E[U(x)] = U(\bar{x})$$

$$CE = \bar{x}$$

$$0 = A$$



Risk loving individual

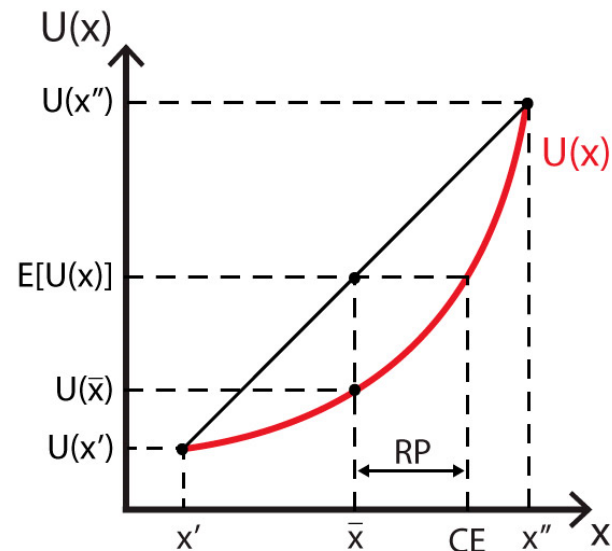
$$E[U(x)] > U(\bar{x})$$

$$CE > \bar{x}$$

$$0 > A$$

Risk Preferences

Those valuing larger sums more than smaller sums will seek to maximize their risk spread and thus are called risk-loving.



Risk loving individual

$$E[U(x)] > U(\bar{x})$$

$$CE > \bar{x}$$

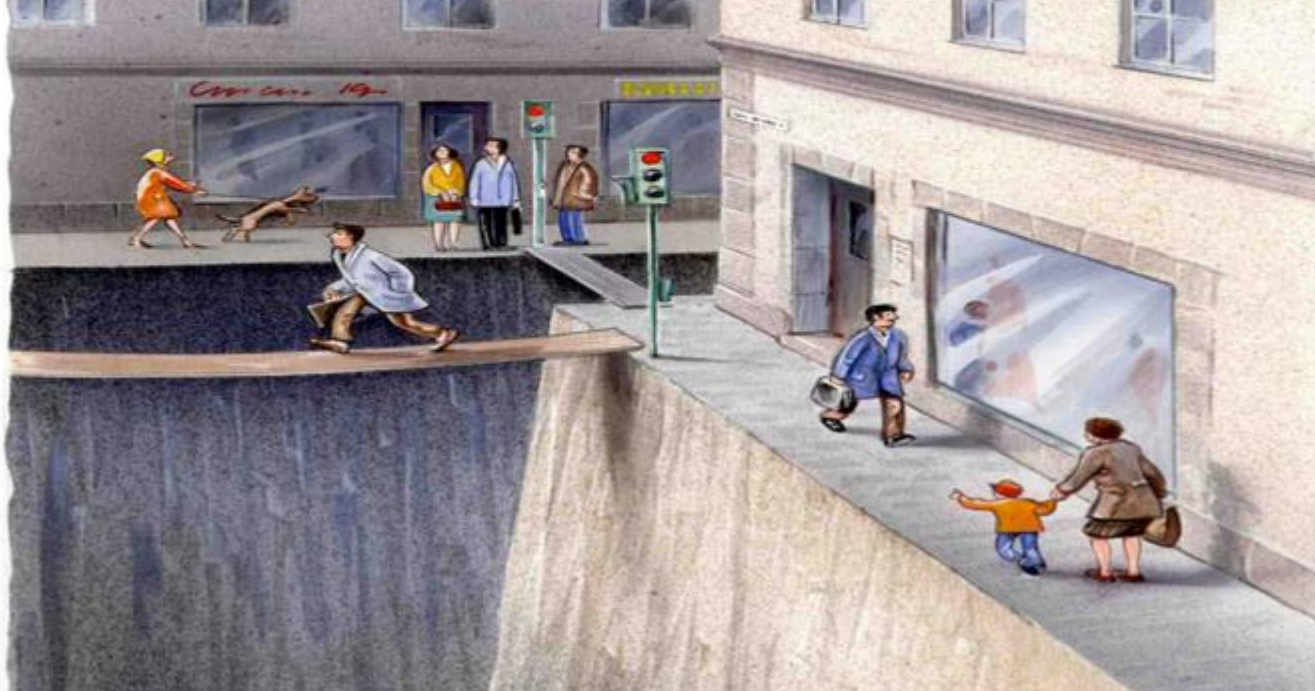
$$0 > A$$

Subjective evaluations of risk

Including risk preferences into expected value calculations make them subjective: recording how different people value risk.

But why stop at these specific risk preferences?





Very different feeling – yet objectively, in terms of likely injuries comparable:

Tempo 30 Unfall Fallhöhe von circa 3,5 Meter

Tempo 50 rund 9,5 Meter

Tempo 70 sogar 19 Meter



Psychological findings:

- Involuntary risks are seen as worse than voluntary risks.
- Risks where one lacks control are seen as worse.
- Novel risks are seen as worse than accustomed risks.
- Large risks are underestimated (traffic risks); small risks are overestimated (terror attack).
- Small probability and very bad consequences (nuclear power) is seen as worse than high probability and smaller negative consequences.

Psychological findings:

- Involuntary risks are seen as worse than voluntary risks.
- Risks where one lacks control are seen as worse.
- Novel risks are seen as worse than accustomed risks.
- Large risks are underestimated (traffic risks); small risks are overestimated (terror attack).
- Small probability and very bad consequences (nuclear power) is seen as worse than high probability and

Should those subjective evaluations be included in risk evaluations – like risk preferences – or should they be excluded as "irrational biases" or "cognitive mistakes"?

Currently no agreement on this question, nevertheless it is important for the understanding of what risk is or might be!

SUMMARY – quantitative model of risk

- EV model
- arguments for it: econ optimization & fact-value dist.
- measurement of definition? -> other measures: VAR!
- Risk preferences
- Problem of subjective evaluations

3. Deciding under Ignorance & Deep Uncertainty

Determine the risk of climbing Kebnekaise (or some other mountain you know) in early October!



Determine the risk of climbing Kebnekaise (or some other mountain you know) in early October!

In the previous chapter I discussed quantitative models of risk. But they might not be applicable to such a case, because we lack important information. Let's remember my earlier classification:

Decision making under certainty	Action has only one outcome
Decision making under risk	Action has more than one possible outcomes ; results of each outcome is known; probability of each outcome is known .
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seems to be the case here, as in many other real-world decisions...



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focus on these two categories in this section

Under Ignorance - TRANSFORMATION strategies

If there are no probabilities known, assign equal probabilities to all possibilities!

Principle of Insufficient Reason

If there are no probabilities known, assign equal probabilities to all possibilities!

Technology A	200 dead	40 dead	0 dead
Technology B	150 dead	20 dead	10 dead
Technology C	100 dead	100 dead	100 dead

Principle of Insufficient Reason

If there are no probabilities known, **assign equal probabilities** to all possibilities!

	1/3	1/3	1/3
Technology A	200 dead	40 dead	0 dead
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Principle of Insufficient Reason

If there are no probabilities known, assign equal probabilities to all possibilities!

	1/3	1/3	1/3	EV_{PIR}
Technology A	200 dead	40 dead	0 dead	80
Technology B	150 dead	20 dead	10 dead	60
Technology C	100 dead	100 dead	100 dead	100

Principle of Insufficient Reason

If there are no probabilities known, assign equal probabilities to all possibilities!

	1/3	1/3	1/3	EV_{PIR}
Technology A	200 dead	40 dead	0 dead	80
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Technology C	100 dead	100 dead	100 dead	100

PIR – ranking: **B > A > C**

Under Ignorance - AVOIDINGS PROBS strategies

Identify the **worst** outcome of each option and choose that option with the **least worst** outcome!

MaxiMin or MiniMax

Identify the worst outcome of each option and choose that option with the least worst outcome!

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MiniMax – ranking: **C** > **B** > **A**

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MiniMax – ranking: **C > B > A**

PIR – ranking: **B > A > C**

MaxiMin or MiniMax

Identify the worst outcome of each option and choose that option with the least worst outcome!

Technology A	200 dead	40 dead	0 dead
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Technology C	100 dead	100 dead	100 dead

What if not all possibilities are known?

MiniMax – ranking: **C > B > A**

PIR – ranking: **B > A > C**

Safety factors in engineering

- Rules of thumb
- identify intended maximal loads and disturbances of the object/structure
- specify how many times over the object structure is to withstand max intended load = safety factor
- e.g. safety factor of 2 when building a bridge: bridge is designed to resist twice the maximal load to which it will in practice be exposed

Why safety factors?

intended to compensate for five major types of sources of failure:

- (1) Higher loads than those foreseen
- (2) Worse properties of the material than foreseen
- (3) Imperfect theory of the failure mechanism in question
- (4) Possibly unknown failure mechanisms
- (5) Human error (e.g., in design)

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- (5) Human error (e.g., in design)

quantifiable, but perhaps uncertain

-> ignorance; thus use qualitative approaches

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- (1) Higher loads than those foreseen
- (2) Worse properties of the material than foreseen
- (3) Imperfect theory of the failure mechanism in question
- (4) Possibly unknown failure mechanisms
- (5) Human error (e.g., in design)

not quantifiable, as possible outcomes are unknown

-> deep uncertainty, non-probabilizable, requires other approaches

-> ROBUSTNESS: failure avoided for most/all known possibilities, irrespective of their probability and severity of outcomes

-> PRECAUTION: putting burden of proof and responsibility on designer

Precautionary Principle

If an action or policy has a **suspected risk of causing harm** to the public or to the environment, in the **absence of scientific consensus** that the action or policy is harmful, the **burden of proof** that it is *not* harmful falls on **those taking an action**.

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- How small suspicion is sufficient?
- What counts as scientific consensus?
- How does one prove that something is *not* harmful?
- How do we apply it without becoming paralyzed?

Arguments for qualitative risk modelling

- computational costs
- simplicity
- lack of accuracy
 - for many projects, the probabilistic quantifications of the effects of safety measures are generally not considered accurate enough to be used as direct inputs into economic analysis.
- security
 - if accuracy is not sufficiently high, it is preferable to develop a robust structural design rather than an economically optimized one

SUMMARY – qualitative models of risk

- many cases where we lack sufficient information to build/rely on qualitative model
- strategies for ignorance
- strategies for deep uncertainty
- probabilistic risk analysis is **one of several** tools for risk assessment. Not a sure source of final answers !
- give different results, thus have a substantial influence on risk assessment. Justify!

4. Ethical Dimensions of Risk Management

Risk assessment



Risk management

Collecting and assessing the relevant information and on this base characterizing the nature and magnitude of the risk.

deciding which actions to take & who to expose to the risk of their outcomes

Utilitarian approach (recall: version of consequentialism)

actualist approach: assign to an action with uncertain outcomes the utility of the outcome that actually materializes.

e.g. "reinforce a bridge before it is used for a single, very heavy transport?"

50% risk that the bridge will fall down if it is not reinforced

choice: not to reinforce the bridge

result: everything goes well; the bridge is not damaged.

According to the actualist approach, the decision was right. This is, of course, contrary to common moral intuitions.

maximization of expected utility approach: the utility of a mixture of potential outcomes is defined as the probability-weighted average of the utilities of these outcomes.

Utilitarian approach - criticisms

(1) disallows a common form of cautiousness - disproportionate avoidance of large disasters. For example:

- human deaths are valued equally and additively
- EU utilitarian must be indifferent between "probability of 1 in 1000 that one person will die" to "probability of 1 in 100000 that fifty persons will die".
- intuitively, many would say that the second risk is worse, because 50 people might die
- perhaps deal with it by including risk preferences, but that creates problems how to constrain subjective factors, as i discussed.

(2) disallows common striving for fairness - disproportionate avoidance of high-probability risks for particular individuals. E.g.:

- choice between "exposing one person to a probability of 0.9 to be killed" vs. "exposing each of one hundred persons to a probability of 0.01 of being killed"
- EU utilitarian, valuing human deaths equally and additively requires that the former alternative be chosen.

In summary, because insists on proportionality, EU utilitarian prohibits what morally reasonable standpoints on risk taking and risk imposition.

Deontological approach – recall:

- People have certain rights, which must be respected
- Certain intentions are not permissible (e.g. intending to harm or deceive)

Deontological approach

- problem of expanding deontological reasoning into risk management:
“Imposing how slight a probability of a harm that violates someone’s rights also violates his rights?” (Nozick 1974, 74).
- e.g. "if *A* has a right that *B* does not bring about a certain outcome, then *A* also has a right that *B* does not perform any action that (at all) increases the probability of that outcome."
- A has the right not to be killed
- car accidents of type X would kill A
- B, by driving his car in A's town, would increase the probability of a car accident of type X
- Therefore, B must not drive in A's town

Deontological approach

- problem of expanding deontological reasoning into risk management:

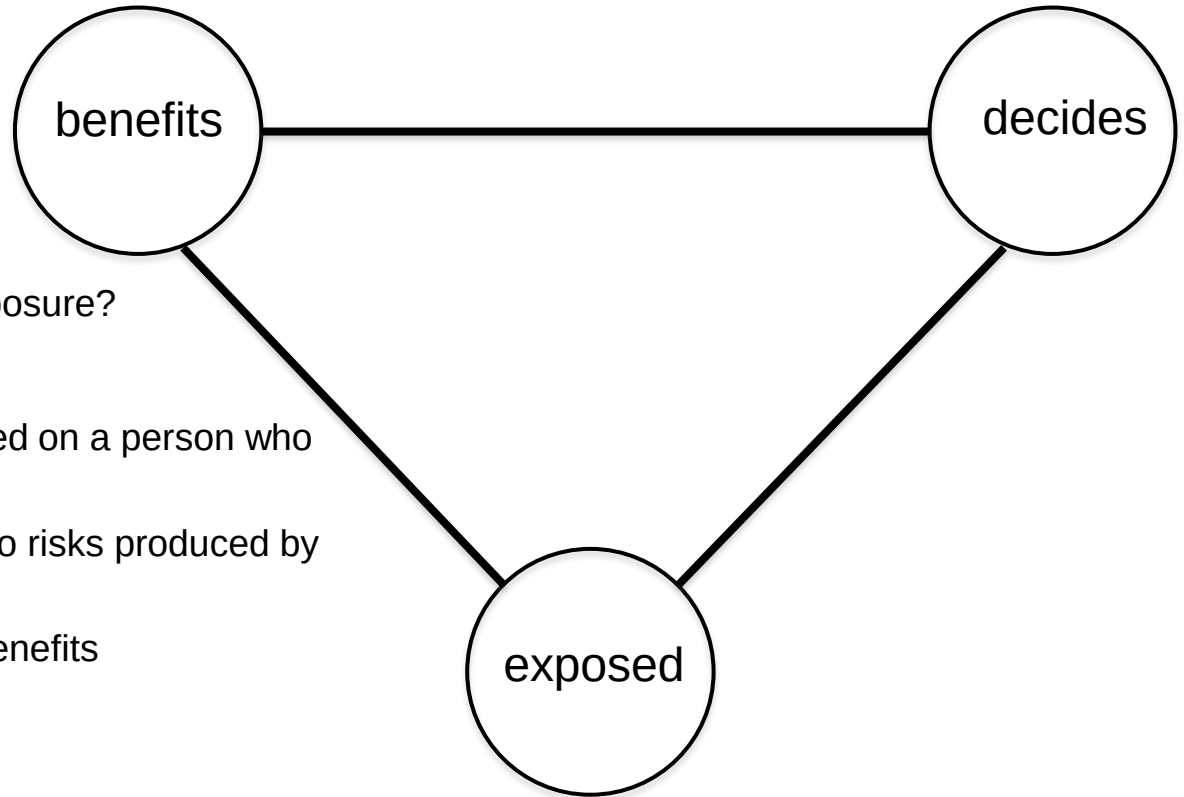
“Imposing how slight a probability of a harm that violates someone’s rights also violates his rights?” (Nozick 1974, 74).

- Thus, sometimes right not to be exposed to risk sometimes be overridden. But under what conditions? Candidates:

- intentional and unintentional risk exposure
- voluntary risk-taking vs. risks imposed on a person who does not accept them.
- exposure to self-produced risks vs. to risks produced by others
- benefits from risk exposure vs. no benefits

This cannot be done in a framework that treats risks as probabilistic mixtures of outcomes.

Conditions between three stakeholders



decision-maker

intentional and unintentional risk exposure?

risk-exposed

voluntary risk-taking vs. risks imposed on a person who does not accept them?

exposure to self-produced risks vs. to risks produced by others?

benefits from risk exposure vs. no benefits

beneficiary

exposed to risk or not?

SUMMARY – ethical aspects of risk management

- Ethical considerations should have influence in risk management!
- utilitarian vs. deontological approach
- mere application of EU for moral purposes not sufficient
- complex considerations of people's rights and how their role as different stakeholders might void these rights